Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/08/2024
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Analysis of ISS Data from the Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/01/2022	**End Date:**	06/30/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2026 per J. McQuillen/GRC (Ed., 9/29/25). NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 10/24/24).		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Hasan, Mohammad Mujibul M.Sc. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC22K0328		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As space missions increase in scope, size, complexity, and duration, so do both power and heat dissipation demands. This is particularly the case for future manned missions to Mars. Paramount to the success of these missions is the ability to reduce size and weight, including those of thermal management sub-systems. One means to achieving this goal is to transition from single-phase to two-phase thermal management. By capitalizing upon the merits of both latent and sensible heat exchange rather than sensible exchange alone, two-phase systems can yield orders of magnitude enhancement in evaporation and condensation heat transfer coefficients compared to single-phase systems. These improvements are evident from recent NASA workshops that culminated in critical recommendations concerning the implementation of flow boiling and condensation in a variety of space applications such as Rankine cycle power conversion, thermal control systems, and advanced life support systems. The Flow Boiling and Condensation Experiment (FBCE) was conceived in 2011 with the intent of developing an integrated two-phase flow boiling/condensation facility for the International Space Station (ISS) to serve as a primary platform for obtaining two-phase flow and heat transfer data in microgravity. By comparing the microgravity data against those obtained in Earth's gravity, it will be possible to ascertain the influence of body force on two-phase transport phenomena in pursuit of predictive design tools, and to help determine minimum flow criteria that would ensure gravity independent flow boiling and condensation. FBCE is a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Key objectives of the proposed project will be to acquire information from the ISS microgravity heat transfer data and video records as well as assess validity and accuracy of recorded data for different operating conditions. The ISS data will be used to (1) assess and retrofit available empirical correlations and demonstrate validity for different gravities, (2) assess and retrofit available theoretical models and demonstrate validity for different gravities, (3) assess and retrofit models for minimum velocity that would ensure gravity independent heat transfer, and (4) develop computational fluid dynamics (CFD) models for both flow boiling and flow condensation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Advanced cooling methods for electric vehicles. 2. Cooling of fusion reactors. 3. Cooling of avionics. 4. Cooling of hybrid vehicle power electronics 5. Design of industrial heat exchangers.
Task Progress:	This study is an elaboration on flow instabilities observed during flow boiling experiments conducted aboard the International Space Station (ISS) as part of the Flow Boiling and Condensation Experiment (FBCE). During highly subcooled flow boiling, liquid backflow into the channel that rapidly condenses vapor within the channel was observed. Experiments were conducted with an enhanced sampling frequency of 30 Hz and an extended image sequence recording duration of at least 4 seconds at 500 frames per second to further investigate the instability. Identical experiments were performed both in microgravity aboard the International Space Station (ISS) and in Earth gravity during vertical upflow. Instabilities in microgravity are more severe than those in Earth gravity and, at their most severe, instabilities propagate to the channel's upstream region. Instabilities are observed within the channel when the intensity, I, which is dependent inlet pressure fluctuations and mass velocity, exceeds 1.8×105 W/m2. Parametric trends of the frequency and amplitude of inlet pressure fluctuations during instability are examined, which reveal instabilities are most severe at low flow rates, high inlet subcoolings, and high heat fluxes. Various stability maps proposed in the literature are evaluated against the present database, and instabilities only manifest for subcooling numbers greater than 14. A subset of the database containing the Onset of Flow Instability (OFI) point is extracted to first evaluate correlations available in the literature and then develop a new correlation. The new correlation is applicable in both microgravity and Earth gravity conditions and predicts the database with a Mean Absolute Error (MAE) of 1.3%.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Mudawar I, Darges SJ, Devahdhanush VS. "Prediction technique for flow boiling heat transfer and critical heat flux in both microgravity and Earth gravity via artificial neural networks (ANNs)." International Journal of Heat and Mass Transfer. 2024 Mar 1;220:124998. https://doi.org/10.1016/j.ijheatmasstransfer.2023.124998 , Mar-2024
Articles in Peer-reviewed Journals	Mudawar I, Devahdhanush VS, Darges SJ, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Microgravity flow boiling experiments with liquid-vapor mixture inlet onboard the International Space Station." International Journal of Heat and Mass Transfer. 2024 Jun 1;224:125299. https://doi.org/10.1016/j.ijheatmasstransfer.2024.125299 , Jun-2024
Articles in Peer-reviewed Journals	Huang CN, Chang S, Suh Y, Mudawar I, Won Y, Kharangate CR. "Machine learning boiling prediction: From autonomous vision of flow visualization data to performance parameter theoretical modeling." International Journal of Multiphase Flow. 2024 Sep 1;179:104928. https://doi.org/10.1016/j.ijmultiphaseflow.2024.104928 , Sep-2024
Articles in Peer-reviewed Journals	Afzal A, Lee S, Kim SM, Mudawar I. "Gaussian process regression to predict dryout incipience quality of saturated flow boiling in mini/micro-channels." Applied Thermal Engineering. 2024 Nov 1;256:124137. https://doi.org/10.1016/j.applthermaleng.2024.124137 , Nov-2024
Articles in Peer-reviewed Journals	Noh H, Lee S, Kim SM, Mudawar I. "Utilization of XGBoost algorithm to predict dryout incipience quality for saturated flow boiling in mini/micro-channels." International Journal of Heat and Mass Transfer. 2024 Oct 1;231:125827. https://doi.org/10.1016/j.ijheatmasstransfer.2024.125827 , Oct-2024
Articles in Peer-reviewed Journals	Mudawar I, Darges SJ, Devahdhanush VS. "Critical heat flux for flow boiling with saturated inlet in microgravity onboard the International Space Station." International Journal of Heat and Mass Transfer. 2024 Nov 15;233:126017. https://doi.org/10.1016/j.ijheatmasstransfer.2024.126017 , Nov-2024
Articles in Peer-reviewed Journals	Mudawar I, Darges SJ, Devahdhanush VS, Hasan MM, Nahra HK, Balasubramaniam R, Mackey JR. "Two-Phase Flow Instabilities during Microgravity Flow Boiling onboard the International Space Station, Critical heat flux for flow boiling with saturated inlet in microgravity onboard the International Space Station." International Journal of Heat and Mass Transfer. 2024 Dec 1;234:126102. https://doi.org/10.1016/j.ijheatmasstransfer.2024.126102 , Dec-2024